Assignment 2: Lane detection

Introduction

The scope of this assignment is to develop a lane detection algorithm that works with real images from PandaSet data set, available at:

https://www.kaggle.com/datasets/usharengaraju/pandaset-dataset.

The archive with the images to be used is available at the following link (17.26 GB):

https://www.dropbox.com/scl/fi/sch1t7ns9vfpa22setcfw/archive.zip?rlkey=0sglcvm9l9xbzb81zoerkhx0b&dl=0

The camera parameters are as follows:

```
camera = struct(
        'ImageSize',[1920 1080],'PrincipalPoint',[970 483],...
         'FocalLength',[1970 1970],'Position',[1.8750 0 1.6600],...
         'Rotation',[0 0 0]
);
focalLength    = camera.FocalLength;
principalPoint = camera.PrincipalPoint;
imageSize      = camera.ImageSize;
height         = camera.Position(3);
pitch          = camera.Rotation(2);
```

Assignment

1. Develop an implementation either using MATLAB or Python 3 of the GOLD lane detection algorithm.
2. The code shall receive as input the directory containing the images to be processed (e.g., PandaSetSensorData/archive/008/Camera/front_camera/)
3. (**mandatory requirement**) The code shall process each image in the path, and produce the following output:

a. The image shall be displayed, and the detected lane delimiter(s) shall be displayed as green vertical line(s) over the bird's eye view of the input image, as shown below:

b. In case both the lanes are not found, a message shall be written on the bird's eye view, such as for example "No lanes found".

4. (**optional requirement 1**) The code shall be able to discriminate between continuous lane delimiters, and dashed lane delimiters.
5. (**optional requirement 2**) The visualization of the lane delimiter shall be performed on the original image instead of the bird's eye view.
6. **(optional requirement 3)** The algorithm shall recognize the presence of obstacles in the lane and tell the approximate distance.

Instructions

- The report is individual.
- Upload you work as a ZIP file using Portale della didattica.
- The ZIP file must contain:
  - A PDF file with a report describing the algorithm implemented.
  - MATLAB/Python code you implemented.
- The code will be tested against the PandaSet data set front_camera
  - In case of MATLAB implementation: a variable named **search_path** shall be used to point to the directory containing the images to be processed, as in the following example:

```
search_path = 'PandaSetSensorData/archive/044/Camera/front_camera/*.jpg';
```

- In case of Python implementation: the python script shall be named **run_gold.py**, and the search path of the directory containing the images to be processed shall be provided as argumento on the command line, as in the following example:

```
python3 run_gold 'PandaSetSensorData/archive/044/Camera/front_camera/*.jpg
```

- The deadline for delivering the assignment is: April 21st, 2026 at 11:30.